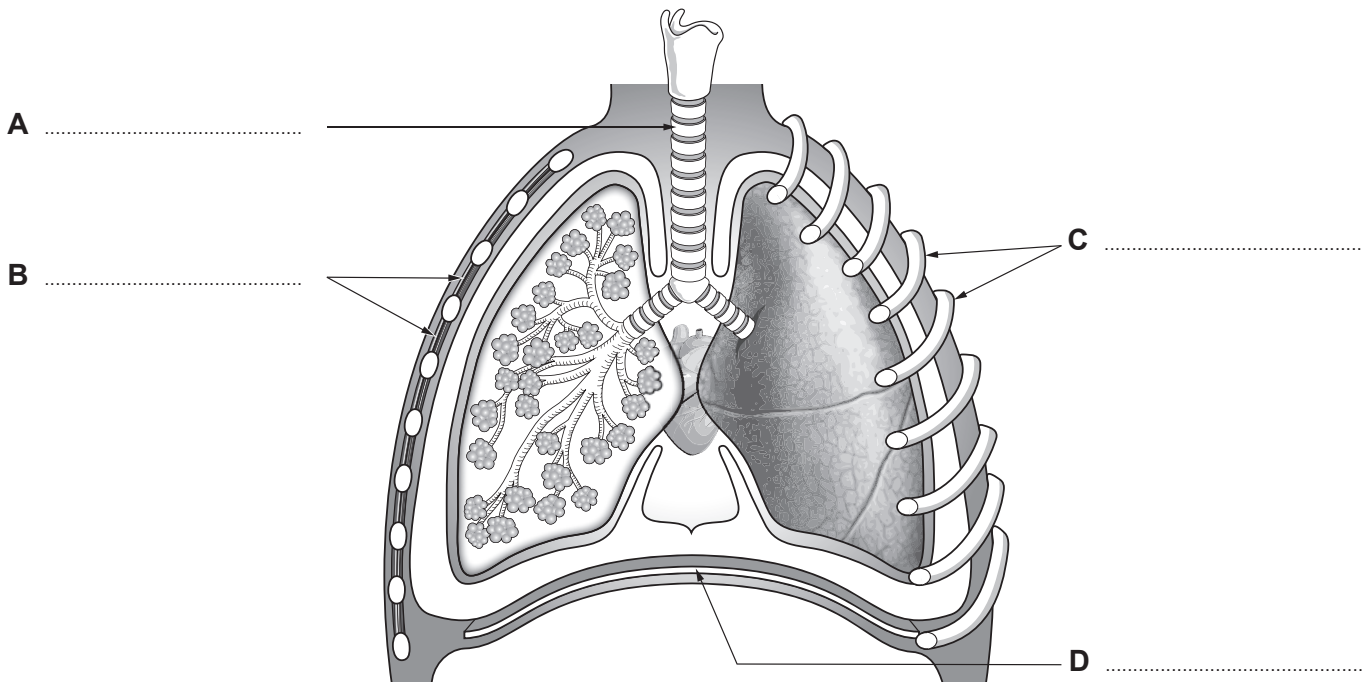


5. Sports science students are learning about the respiratory system.

(a) (i) Label the diagram below using words from the box.

[4]

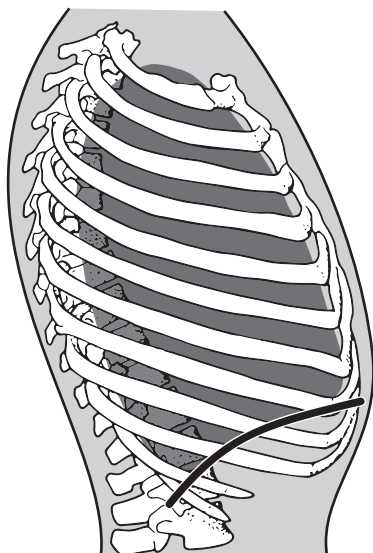
ribs intercostal muscles diaphragm lung trachea



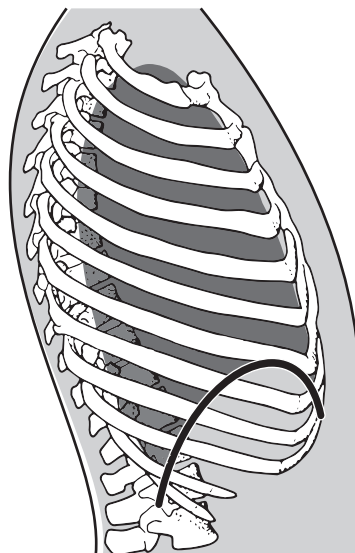
Examiner
only

- (ii) Use your knowledge and the diagrams below to describe how the changes in the diaphragm, ribcage and intercostal muscles allow you to breathe in and out.

[6 QER]



breathing in

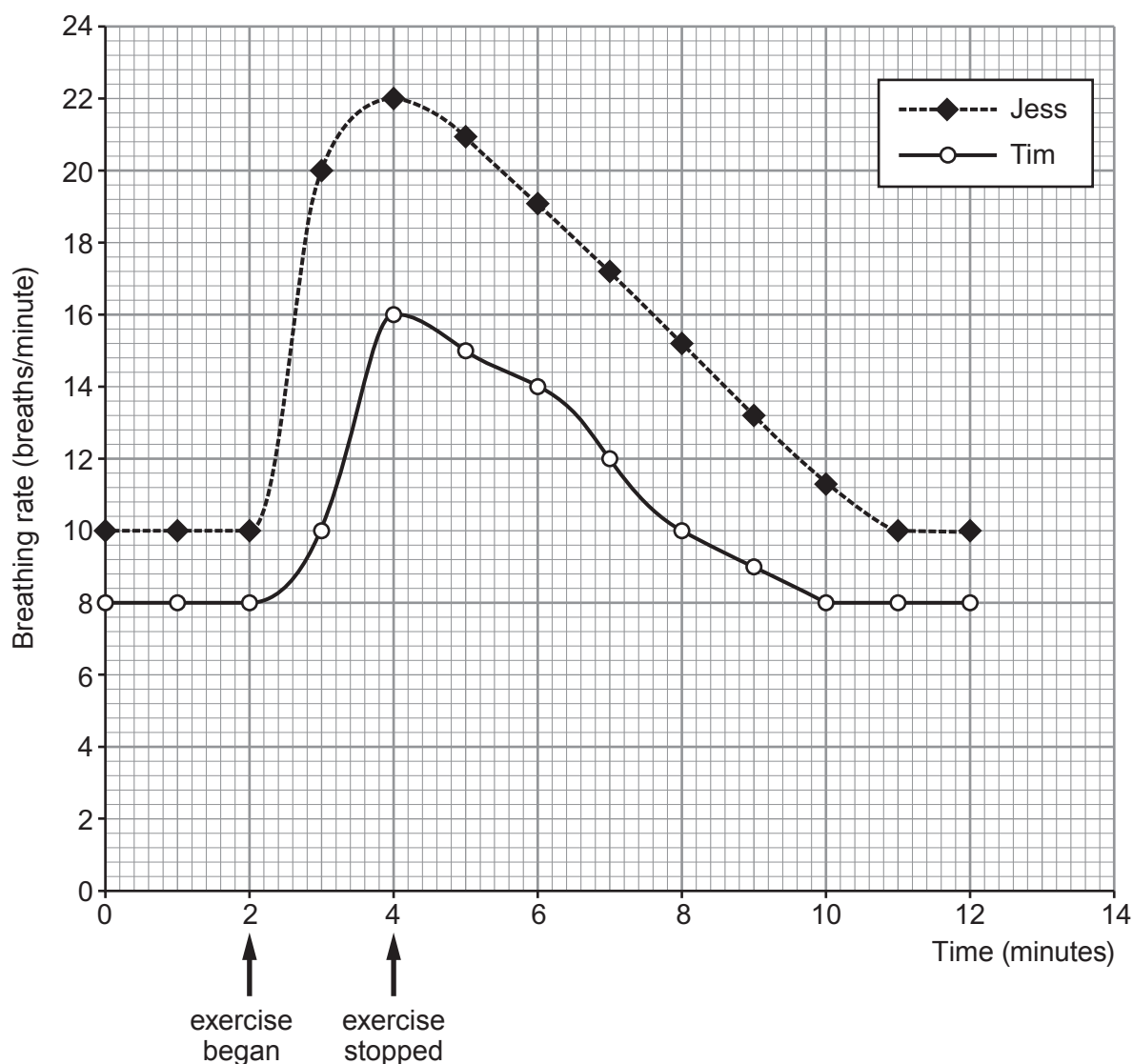


breathing out

- (b) Two students, Tim and Jess, studied the effect of exercise on breathing rate. They followed the method below:

1. record breathing rate before exercising
2. exercise for 2 minutes by running on a treadmill
3. measure the breathing rates during and after exercise up until 12 minutes

Their results are shown on the graph below.



Use the graph to complete the following sentences.

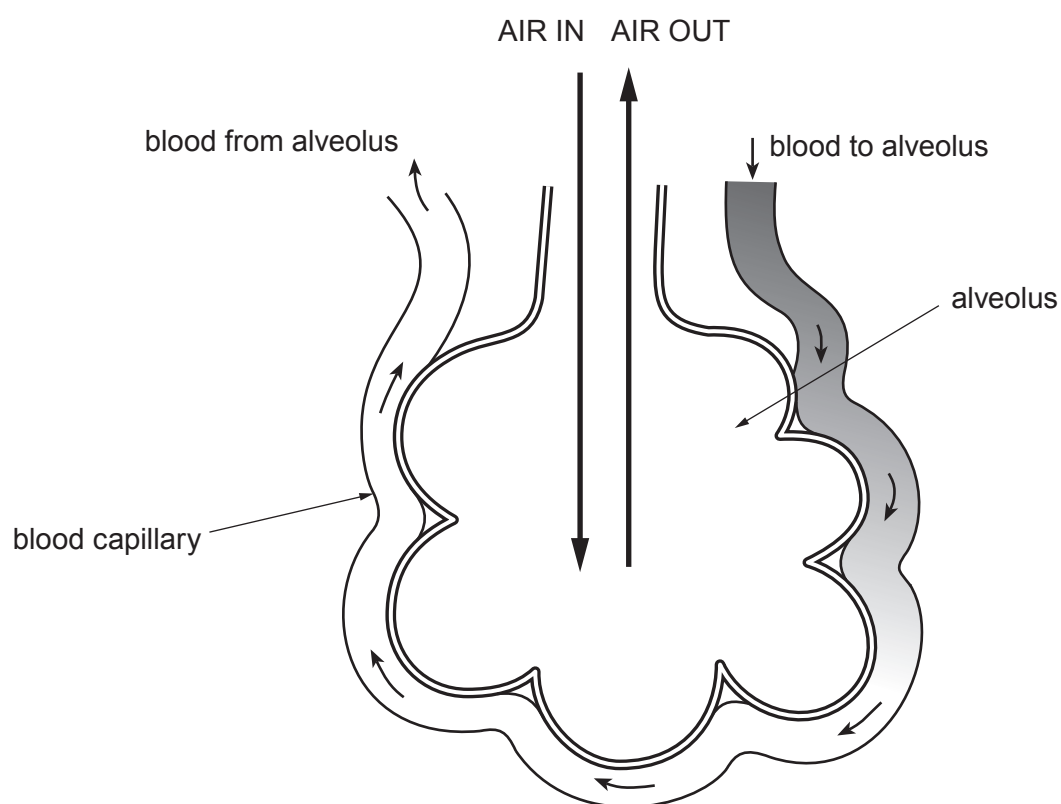
[3]

At 2 minutes, the difference between Tim and Jess' breathing rate is breaths per minute.

Between 8 and 10 minutes Tim's breathing rate drops than Jess' breathing rate.

Tim's breathing rate returns to normal at minutes.

(c) The diagram below is of an alveolus within the lung where gases are exchanged.



(i) State which gases are exchanged.

[2]

..... and

(ii) Name the process by which these gases move between the alveolus and the capillary.

[1]

.....

(iii) State **two** ways in which the alveoli are adapted for gas exchange.

[2]

1.

2.